

NAME

aed — perform aes256-cbc encryption/decryption

SYNOPSIS

aed [**-deh**]

DESCRIPTION

The **aed** utility can be used to perform symmetric encryption/decryption of the input stream using 256bit AES in Cipher Block Chaining mode with a SHA256 digest.

OPTIONS

aed supports the following command-line options:

- d** Perform decryption of the input stream.
- e** Perform encryption of the input stream.
- h** Print a short usage and exit.

DETAILS

aed reads data from stdin and either encrypts or decrypts it (depending on the **-d** or **-e** flag). It uses 256bit AES CBC mode with a SHA256 digest with keying material derived from the passphrase using the `EVP_BytesToKey(3)` function, generating a suitable salt via `RAND_bytes(3)`.

aed reads the password from which to derive the key material from the `AED_PASS` environment variable.

When encrypting, the ciphertext output is prefixed by the string "Salted__", followed by the 8 byte salt. Output is written to stdout.

EXAMPLES

To encrypt the contents of the file 'file' and storing the encrypted output in 'file.enc':

```
aed -e <file >file.enc
```

To decrypt the contents of that file again:

```
aed -d <file.enc
```

Since **aed** operates on stdin and stdout, the above two commands could also be chained:

```
cat file | aed -e | aed -d
```

This program is compatible with the following `openssl` invocation:

```
openssl enc -aes-256-cbc -md sha256 -pass env:AED_PASS
```

EXIT STATUS

aed exits 0 on success, and >0 if an error occurred.

SEE ALSO

`EVP_BytesToKey(3)`, `EVP_EncryptInit(3)`, `RAND_bytes`

<https://www.netmeister.org/blog/passing-passwords.html>

https://wiki.openssl.org/index.php/EVP_Symmetric_Encryption_and_Decryption

HISTORY

This program (or variants thereof) was first assigned as a stand-alone programming assignment for the class “Advanced Programming in the UNIX Environment” at Stevens Institute of Technology in the Fall of 2012.